Conditional Generative Drifting for One-Step One-to-Many Super-Resolution

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Abstract

This paper investigates the applicability of the Generative Drifting framework to one-step single-image super-resolution in the one-to-many setting. Since a single low-resolution image may correspond to multiple plausible high-resolution reconstructions, super-resolution is inherently ill-posed; however, most computationally efficient methods still produce averaged and over-smoothed results. To address this limitation, we extend the Generative Drifting paradigm to conditional super-resolution and propose a stochastic one-step formulation that learns a direct mapping from a low-resolution input and random noise to plausible high-resolution outputs in a single forward pass. The proposed method performs drifting in a residual feature space, encouraging generated samples to move toward plausible high-resolution targets while mitigating collapse to a single solution. Experimental results show that, compared with a noise-conditioned baseline trained with pixel-wise, perceptual, and low-resolution consistency losses, the proposed approach yields more diverse and perceptually plausible reconstructions and better captures the one-to-many nature of the super-resolution problem, without compromising PSNR and LPIPS performance.

1 Introduction

Single-image super-resolution (SR) aims to reconstruct a high-resolution (HR) image from a low-resolution (LR) observation. Although SR is often formulated as a supervised image-to-image regression problem, it is fundamentally ill-posed: the degradation process removes high-frequency information, and therefore multiple HR images may be consistent with the same LR input [1–3]. This ambiguity is especially pronounced in textured regions, where different reconstructions may preserve the same low-frequency structure while differing in fine details. Consequently, SR should be viewed not only as a deterministic restoration task, but also as a conditional generative problem.

Classical deep SR methods such as SRCNN [4], VDSR [5], EDSR [6], and RCAN [7] achieve strong distortion-oriented performance under paired supervision. However, when trained with pixel-wise objectives, they are encouraged to predict a single conditional mean and therefore often produce averaged and over-smoothed textures [2, 8, 9]. Perceptual and adversarial approaches, including SRGAN [8], ESRGAN [9], and Real-ESRGAN [10], improve visual sharpness and perceptual realism, but are also associated with unstable optimization, hallucinated high-frequency details, and limited control over the set of admissible outputs [9, 10]. Thus, existing SR methods often face a familiar trade-off: distortion-oriented models are faithful but over-smooth, whereas perceptual models are sharper but less stable and less controllable.

A more principled direction is to treat SR explicitly as conditional generation. SRFlow [1] models the conditional distribution of HR images given an LR input rather than a single point estimate. Adaptive Target Generation [2] further argues that the paired HR image should not always be interpreted as the only valid supervisory target. Related ideas also appear in one-to-many SR formulations, where several plausible HR outputs are generated for the same LR input [3]. These works highlight that SR ambiguity is not a training artifact but a structural property of the problem.

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Another relevant line of research concerns sample construction and representation geometry in low-level restoration. In particular, contrastive SR methods show that naive positive-negative construction is insufficient for dense prediction tasks. PCL-SR [11] demonstrates that sharpened HR variants can serve as informative positives and mildly blurred HR variants as hard negatives, since SR requires feature comparisons that are sensitive to texture and degradation rather than to global semantic similarity. This observation is directly relevant in the one-to-many setting, where one must specify not only how to generate diverse outputs but also what kinds of alternatives should be regarded as plausible.

In parallel, generative modeling has increasingly moved toward efficient one-step generation. Diffusion-based SR methods such as SR3 [12] and latent restoration approaches based on diffusion priors [13] produce highly realistic outputs, but their iterative inference is computationally expensive. Considerable effort has therefore been devoted to acceleration, including progressive distillation [14], consistency-type approximations [15], flow-based one-step models [16, 17], and fast SR-oriented approximations such as SinSR [18] and OSEDiff [19]. These methods reduce inference cost, yet most remain conceptually tied to diffusion or flow trajectories, teacher models, or distillation pipelines.

A recent alternative is Generative Drifting [20], which reformulates generation as the evolution of the pushforward distribution during training rather than at inference time. In that framework, a generator is trained through a drifting field and a fixed-point regression objective toward a frozen drifted target [20]. The original method already supports feature-space drift and conditional generation [20]. However, super-resolution poses a substantially different form of conditioning. In class-conditional generation, the condition specifies a semantic category while leaving the instance largely unconstrained. In SR, by contrast, the condition is an observed LR image that imposes a strong measurement constraint on the output. This motivates the present work: we investigate whether the fixed-point drifting principle can be adapted to one-step conditional super-resolution in the one-to-many setting. Our central idea is to specialize drifting to the geometry of SR, so that generated samples move toward admissible HR structures associated with the same LR observation, while being discouraged from collapsing to identical reconstructions. In this way, our goal is to preserve the efficiency of one-step generation while enforcing fidelity to the measurement and allowing controlled diversity only within the admissible solution set.

The main contributions of this work are as follows:

- We cast single-image super-resolution as a conditional one-to-many generation problem and extend the Generative Drifting framework to the observation-conditioned SR setting, where the low-resolution image itself serves as the condition.
- We propose an SR-oriented drifting formulation in residual feature space that combines attraction to admissible high-resolution targets with repulsion between samples generated from the same low-resolution input, enabling controlled diversity without collapse.
- We demonstrate that the proposed one-step stochastic SR framework produces more diverse and perceptually plausible reconstructions, while maintaining competitive reconstruction quality relative to a noise-conditioned baseline trained with pixel-wise, perceptual, and low-resolution consistency losses.

2 Problem Statement

Setting. We consider single-image super-resolution (SISR) with a fixed upscaling factor r. Given a low-resolution image $x^{LR} \in \mathbb{R}^{h\times w\times c}$, the goal is to reconstruct a high-resolution image $x^{HR} \in \mathbb{R}^{H\times W\times c}$, where $H = rh$ and $W = rw$. In the standard synthetic setting, the LR image is obtained by bicubic degradation,

$$x^{LR} = D(x^{HR}),$$

where D denotes bicubic downsampling.

Training data and ambiguity. Let $\mathcal{D} = \{(x_i^{LR}, x_i^{HR})\}_{i=1}^N$ be a paired training set. We assume that the conditional distribution $p(x^{HR} \mid x^{LR})$ is non-degenerate: a single LR observation may correspond to several plausible HR reconstructions that share the same low-frequency structure while

differing in fine high-frequency details. Accordingly, we treat SISR as a one-to-many conditional generation problem rather than as a purely deterministic regression task.

Stochastic generator. To model this ambiguity, we use a stochastic generator G_θ parameterized by θ. Given an LR image and an auxiliary noise variable $\varepsilon \sim p_\varepsilon$, the model produces an HR candidate:

$$\hat{x}_i^{(k)} = G_\theta(x_i^{LR}, \varepsilon_i^{(k)}), \qquad k = 1, \ldots, K.$$

Thus, for a fixed input x_i^{LR}, repeated sampling of ε yields multiple candidate HR reconstructions.

Residual parameterization. Following the residual view of SR, we decompose each prediction into a deterministic bicubic component and a learned correction. Let $u(\cdot)$ denote bicubic upsampling and define $x_i^{UP} = u(x_i^{LR})$. The generated sample is then written as

$$\hat{x}_i^{(k)} = x_i^{UP} + r_i^{(k)},$$

where $r_i^{(k)}$ is the predicted residual. This parameterization separates the low-frequency content, which is largely determined by the LR input, from the ambiguous high-frequency details.

Admissible target set. For each training pair (x_i^{LR}, x_i^{HR}), we define a finite set of admissible HR targets $\mathcal{P}_i = \{x_{i,m}^+\}_{m=1}^M$. This set contains the ground-truth image and additional HR variants that remain consistent with the same LR observation, i.e., $\|D(x_{i,m}^+) - x_i^{LR}\|_1 \leq \delta$. These targets represent plausible HR realizations associated with the same LR input and define the manifold toward which generation should drift.

Training objective. The model is trained with a composite objective

$$L(\theta) = \lambda_{\text{rec}} L_{\text{rec}}(\theta) + \lambda_{\text{LR}} L_{\text{LR}}(\theta) + \lambda_{\text{drift}} L_{\text{drift}}(\theta).$$

Here L_{rec} anchors the prediction to the reference HR target, L_{LR} enforces fidelity to the low-resolution measurement, and L_{drift} is a conditional fixed-point drifting loss, detailed in Section 3.

Goal. Our goal is to learn parameters θ^* such that the resulting model admits one-step inference, preserves measurement consistency, and produces diverse yet plausible reconstructions for the same input.

3 Theoretical Part

3.1 Preliminaries: fixed-point view of drifting

Generative Drifting interprets learning as the evolution of the pushforward distribution during training rather than as an iterative refinement procedure at inference time. Let a generator f_θ map a random variable $\varepsilon \sim p_\varepsilon$ to an output $x = f_\theta(\varepsilon)$, inducing a pushforward distribution

$$q_\theta = [f_\theta]_\# p_\varepsilon.$$

A drifting field $V_{p,q}(x)$ specifies how a generated sample should move so that the current pushforward distribution q evolves toward a target distribution p. In the original Drifting formulation, the key equilibrium property is that when the generated distribution matches the target one, the drifting field vanishes, i.e., $V_{p,q}(x) = 0$. This motivates a fixed-point view of learning: at the optimum, a generated sample should coincide with its drifted version.

Accordingly, the optimal generator satisfies the fixed-point relation

$$f_{\hat{\theta}}(\varepsilon) = f_{\hat{\theta}}(\varepsilon) + V_{p,q_{\hat{\theta}}}\left(f_{\hat{\theta}}(\varepsilon)\right).$$

In practice, this idea is implemented by regressing the current prediction toward a frozen drifted target rather than by directly backpropagating through the drifting field. This yields the fixed-point regression objective

$$\mathcal{L}_{\text{fp}} = \mathbb{E}_\varepsilon \left[\|f_\theta(\varepsilon) - \text{stopgrad}(f_\theta(\varepsilon) + V_{p,q_\theta}(f_\theta(\varepsilon)))\|_2^2 \right],$$

where $\text{stopgrad}(\cdot)$ denotes the stop-gradient operator. Thus, training encourages each generated sample to move toward its drifted version while treating the drift itself as fixed within the current iteration.

3.2 Conditional generative drifting for one-to-many super-resolution

We now adapt the fixed-point drifting principle to conditional one-to-many super-resolution, where a single low-resolution observation x_i^{LR} may correspond to multiple admissible high-resolution reconstructions. In contrast to the unconditional setting of the original Drifting framework, the target distribution is now conditional on the observed LR image itself, which imposes a strong measurement constraint on the output.

Our generator is a stochastic conditional residual model

$$\hat{x}_i^{(k)} = G_\theta(x_i^{LR}, \varepsilon_i^{(k)}), \qquad \varepsilon_i^{(k)} \sim p_\varepsilon,$$

which induces the conditional pushforward distribution

$$q_\theta(\,\cdot \mid x_i^{LR}) = \big[G_\theta(x_i^{LR}, \cdot)\big]_{\#} p_\varepsilon.$$

The goal is therefore not to match an unconditional data distribution, but to evolve the conditional pushforward $q_\theta(\cdot \mid x_i^{LR})$ toward the set of admissible HR reconstructions associated with the same LR observation. Following the original fixed-point logic, we do not directly optimize the drifting field itself. Instead, we construct conditional drift targets in feature space and regress the current prediction toward these frozen targets. In our SR setting, the drifting mechanism is additionally combined with reconstruction and LR-consistency constraints so that diversity is allowed only within the admissible solution set for a fixed observation.

3.3 Residual feature family

We compute drifting in a residual feature space defined relative to the bicubic baseline

$$x_i^{UP} = u(x_i^{LR}),$$

so that the objective focuses on uncertain high-frequency details rather than on the low-frequency structure already explained by bicubic upsampling.

Let $\Phi = \{\phi_j\}_{j=1}^{J}$ denote a family of feature descriptors extracted by a fixed encoder from multiple scales and spatial locations. For each generated sample $\hat{x}_i^{(k)}$ and admissible positive target $x_{i,m}^{+}$, we define residual feature vectors relative to the bicubic baseline:

$$g_{i,j}^{(k)} = \phi_j(\hat{x}_i^{(k)}) - \phi_j(x_i^{UP}), \qquad p_{i,j}^{(m)} = \phi_j(x_{i,m}^{+}) - \phi_j(x_i^{UP}).$$

Thus, drifting is performed on residual feature components that represent plausible high-frequency corrections to the bicubic reconstruction rather than on the full image representation. Full details of the feature family Φ are given in Appendix A.1.

3.4 Admissible positive targets

Unlike the unconditional setting, the positive set is conditional on the LR observation. For each condition i, we define

$$\mathcal{P}_i = \{x_{i,m}^{+}\}_{m=1}^{M}, \qquad x_{i,1}^{+} = x_i^{HR}.$$

For $m \geq 2$, additional admissible targets are constructed from the paired HR image by mild detail-preserving transformations:

$$x_{i,m}^{+} = \mathcal{S}_{\alpha,\sigma}(x_i^{HR}) \qquad \text{or} \qquad x_{i,m}^{+} = \mathcal{H}_\beta(x_i^{HR}),$$

where $\mathcal{S}_{\alpha,\sigma}$ denotes an unsharp-mask type sharpening operator with strength α and blur scale σ, and $\mathcal{H}_\beta$ denotes mild high-pass enhancement with strength β.

A candidate target is accepted only if it remains consistent with the LR observation:

$$\|D(x_{i,m}^{+}) - x_i^{LR}\|_1 \leq \delta.$$

Otherwise, it is replaced by the paired HR image.

3.5 Conditional drifting field

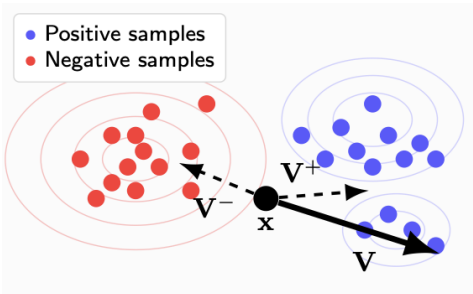


Figure 1: Illustration of the conditional drifting mechanism.

For clarity, in this subsection we fix a condition i and a feature map j, and omit these indices when no ambiguity arises. For each generated residual descriptor $g^{(k)}$, we define the conditional drifting field

$$V^{(k)} = V^{\mathrm{pos},(k)} - V^{\mathrm{same},(k)}.$$

The attraction term is computed over the admissible positive set

$$\mathcal{P} = \{p^{(m)}\}_{m=1}^{M},$$

as

$$V^{\mathrm{pos},(k)} = \lambda_{\mathrm{pos}} \sum_{m=1}^{M} a_m^{(k)} \big(p^{(m)} - g^{(k)}\big),$$

where

$$a_m^{(k)} \propto \exp\left(-\frac{\|g^{(k)} - p^{(m)}\|_2}{\tau_{\mathrm{pos}}}\right), \qquad \sum_{m=1}^{M} a_m^{(k)} = 1.$$

The repulsion term is computed over the alternative generated samples associated with the same LR input,

$$\mathcal{N}^{\mathrm{same},(k)} = \{g^{(t)}\}_{t \neq k},$$

as

$$V^{\mathrm{same},(k)} = \lambda_{\mathrm{same}} \sum_{\substack{t=1 \\ t \neq k}}^{K} b_t^{(k)} \big(g^{(t)} - g^{(k)}\big),$$

where

$$b_t^{(k)} \propto \exp\left(-\frac{\|g^{(k)} - g^{(t)}\|_2}{\tau_{\mathrm{same}}}\right), \qquad \sum_{\substack{t=1 \\ t \neq k}}^{K} b_t^{(k)} = 1.$$

Thus, each generated sample is encouraged to move toward admissible HR residuals while remaining separated from alternative stochastic reconstructions of the same LR observation. Optionally, the repulsion term may be augmented with auxiliary negatives; their detailed construction is given in Appendix A.2.

3.6 Per-condition feature normalization

Again, we fix a condition i and a feature map j, and suppress these indices for readability. Because drifting is defined condition-wise, the normalization should reflect the residual geometry associated with a fixed LR observation. At the same time, all local descriptors extracted from one feature map should share a common scale.

Let $\mathcal{R}^{\mathrm{gen}}$ and $\mathcal{R}^{\mathrm{pos}}$ denote the sets of generated and positive residual descriptors obtained by concatenating all spatial locations of the current feature map. We define the shared feature normalization scale

$$S = \frac{1}{\sqrt{C_j}} \, \mathbb{E}_{u \in \mathcal{R}^{\mathrm{gen}},\, v \in \mathcal{R}^{\mathrm{pos}}} \|u - v\|_2.$$

Residual descriptors are then normalized as

$$\tilde{u} = \frac{u}{S + \varepsilon}, \qquad \tilde{v} = \frac{v}{S + \varepsilon}.$$

The conditional drifting field is computed from these normalized descriptors. Denoting the resulting drift vectors by $\bar{V}$, we further normalize their magnitude by

$$\Lambda = \left(\frac{1}{C_j} \, \mathbb{E}_{\bar{V}} \|\bar{V}\|_2^2\right)^{1/2}.$$

The final normalized drift is

$$\tilde{V} = \frac{\bar{V}}{\Lambda + \varepsilon}.$$

Thus, all local descriptors from the same feature map are compared in a common scale, while the normalization remains specific to the current LR condition.

3.7 Frozen drift target and drifting loss

Following the fixed-point principle of Generative Drifting, we do not backpropagate through the drifting field itself. For a fixed condition and feature map, the drifting loss regresses each normalized residual descriptor toward its drifted version while treating that target as frozen:

$$\ell_{\text{drift}} = \left\| \tilde{g}^{(k)} - \text{stopgrad}\Big(\tilde{g}^{(k)} + \eta\, \tilde{V}^{(k)} \Big) \right\|_2^2.$$

Aggregating over conditions, stochastic samples, and feature maps gives

$$L_{\text{drift}} = \frac{1}{NK} \sum_{i=1}^{N} \sum_{k=1}^{K} \sum_{j=1}^{J} \ell_{\text{drift}}^{i,j,k}.$$

3.8 Training objective

The full training objective combines anchor reconstruction, LR consistency, and conditional drifting:

$$L(\theta) = \lambda_{\text{rec}} L_{\text{rec}} + \lambda_{\text{LR}} L_{\text{LR}} + \lambda_{\text{drift}} L_{\text{drift}}.$$

The anchor reconstruction term is

$$L_{\text{rec}} = \frac{1}{N} \sum_{i=1}^{N} \left\| \hat{x}_i^{(a_i)} - x_i^{HR} \right\|_1, \qquad a_i \sim \text{Unif}\{1, \ldots, K\}.$$

The LR-consistency term is

$$L_{\text{LR}} = \frac{1}{NK} \sum_{i=1}^{N} \sum_{k=1}^{K} \| D(\hat{x}_i^{(k)}) - x_i^{LR} \|_1,$$

which enforces consistency of every stochastic output with the observed LR image.

Together, these terms ensure that one sample remains anchored to the paired HR target, all generated outputs stay faithful to the LR measurement, and residual features evolve toward admissible condition-specific configurations through the fixed-point drifting loss.

3.9 Algorithm

Algorithm 1 Training step of conditional generative drifting for one-step super-resolution

Require: mini-batch $\{(x_i^{LR}, x_i^{HR})\}_{i=1}^{B}$, generator G_θ, degradation operator D, upsampler u, feature family $\Phi = \{\phi_j\}_{j=1}^{J}$

for $i = 1, \ldots, B$ **do**
  compute bicubic baseline $x_i^{UP} \leftarrow u(x_i^{LR})$
  construct admissible positive set $\mathcal{P}_i = \{x_{i,m}^{+}\}_{m=1}^{M}$
  for $k = 1, \ldots, K$ **do**
    sample noise $\varepsilon_i^{(k)} \sim p_\varepsilon$
    generate $\hat{x}_i^{(k)} \leftarrow G_\theta(x_i^{LR}, \varepsilon_i^{(k)})$
  end for
end for
for each condition i and feature map j **do**
  extract residual descriptor matrices from generated and admissible positive samples, relative to x_i^{UP}
  compute shared normalization scales $S_{i,j}$ and $\Lambda_{i,j}$ over all spatial descriptors of feature map j
  compute normalized conditional drift for the generated descriptors
end for
compute conditional drifting loss

$$L_{\text{drift}} = \frac{1}{BK} \sum_{i=1}^{B} \sum_{k=1}^{K} \sum_{j=1}^{J} \left\| \tilde{g}_{i,j}^{(k)} - \text{stopgrad}\Big(\tilde{g}_{i,j}^{(k)} + \eta\, \tilde{V}_{i,j}^{(k)} \Big) \right\|_2^2$$

compute anchor reconstruction loss

$$L_{\text{rec}} = \frac{1}{N} \sum_{i=1}^{N} \left\| \hat{x}_i^{(a_i)} - x_i^{HR} \right\|_1, \qquad a_i \sim \text{Unif}\{1, \ldots, K\}$$

compute LR-consistency loss

$$L_{\text{LR}} = \frac{1}{BK} \sum_{i=1}^{B} \sum_{k=1}^{K} \| D(\hat{x}_i^{(k)}) - x_i^{LR} \|_1$$

form

$$L = \lambda_{\text{rec}} L_{\text{rec}} + \lambda_{\text{LR}} L_{\text{LR}} + \lambda_{\text{drift}} L_{\text{drift}}$$

update parameters θ by one optimizer step minimizing L

4 Experiments

4.1 Experimental setup

Dataset and degradation model. We evaluate the proposed method on single-image super-resolution with an upscaling factor of $\times 4$. In the current study, low-resolution images are generated by bicubic downsampling of the corresponding high-resolution images. Training is performed on the DIV2K training split, and validation is conducted on 100 images from the DIV2K validation split.

Generator architecture and noise injection. The generator is a one-step residual super-resolution network with a U-Net-like encoder–decoder structure. Given an LR image x^{LR}, the model first computes its bicubic upsampling x^{UP} and then predicts a residual correction r, producing the final HR estimate as

$$\hat{x} = x^{UP} + r.$$

For stochastic generation, the model is conditioned on two noise pathways: a spatial noise tensor and a global noise embedding. Unless otherwise stated, all learned models in the main comparison use

the same generator architecture, so that the experiments isolate the effect of the training objective rather than the backbone design.

Compared models. Our primary goal is to evaluate the proposed drifting-based model against strong reference methods trained with the same generator backbone. In particular, we compare:

- **Bicubic interpolation**, used as a non-learned reference;
- **Noise-conditioned baseline**, trained with pixel-wise, perceptual, and LR-consistency losses;
- **Proposed drifting model**, trained with anchor reconstruction, LR-consistency, and conditional drifting losses.

For completeness, we additionally report a deterministic variant without stochastic noise in the supplementary comparison.

Default configuration. Unless otherwise stated, the proposed model uses $K = 4$ generated samples per LR input, $M = 4$ admissible positive targets, and drifting losses computed on three feature levels of a fixed encoder. The admissible positive set contains the ground-truth HR image together with several mild sharpened or high-pass-enhanced variants that remain consistent with the same LR observation.

Checkpoint selection. For stochastic models, the main results are reported using the checkpoint selected by the best validation LPIPS. We additionally monitor the checkpoint with the best validation PSNR. Since stochastic perceptual models may continue to evolve after reaching their best perceptual quality, we do not use the last checkpoint as the primary evaluation point.

4.2 Evaluation metrics

We evaluate both *reconstruction fidelity* and *sample diversity*.

Fidelity metrics. We report:

- **PSNR** between the predicted HR image and the reference HR image;
- **LPIPS** between the predicted HR image and the reference HR image;
- **LR-consistency**, computed as the ℓ_1 distance between the observed LR image and the bicubic downsampling of the prediction.

Diversity metrics. For stochastic models, we generate multiple HR predictions for the same LR input using a fixed set of noise seeds and compute:

- **mean pairwise LPIPS** between generated samples;
- **mean pairwise** ℓ_1 distance between generated samples;
- **mean pairwise downsampled** ℓ_1 distance, computed after bicubic downsampling.

The first two metrics quantify perceptual and pixel-level diversity, while the downsampled pairwise distance reflects whether the induced diversity remains compatible with the same LR observation.

4.3 Main comparison

Table 1 reports the main fidelity metrics, while Table 2 reports diversity-related quantities for stochastic models. We separate these two groups of results because the proposed method is not only a restoration model but also a conditional one-to-many generator, so reconstruction quality and inter-sample variation should be analyzed independently.

The main comparison is designed to answer a simple question: does the proposed conditional drifting mechanism improve the one-to-many behavior of the generator while preserving competitive reconstruction fidelity? The bicubic reference establishes a non-learned baseline, and the noise-conditioned baseline shows what can already be achieved by stochastic conditioning without drifting.

Table 1: Main fidelity comparison on DIV2K validation ($\times 4$).

Model	PSNR ↑	LPIPS ↓	LR-cons ↓
Bicubic	26.91	0.3606	XX.XXXX
Noise-conditioned baseline	27.43	0.3326	XX.XXXX
Proposed drifting model	XX.XX	XX.XXXX	XX.XXXX

Table 2: Main diversity comparison on DIV2K validation ($\times 4$). Diversity metrics are reported only for stochastic models.

Model	Pair LPIPS ↑	Pair ℓ_1 ↑	Pair down ℓ_1 ↓
Noise-conditioned baseline	0.02742	0.00254	0.00026
Proposed drifting model	XX.XXXX	XX.XXXX	XX.XXXX

Our preliminary experiments show that stochastic conditioning alone already produces a measurable degree of variation between samples, but this variation remains limited. By contrast, the proposed drifting model substantially increases the separation between samples generated from the same LR input, indicating a stronger one-to-many effect. At the same time, the fidelity metrics remain competitive with the noise-conditioned baseline, suggesting that the additional diversity is not obtained solely by sacrificing reconstruction quality.

4.4 Fidelity–diversity trajectory analysis

To visualize the trade-off between perceptual fidelity and stochastic diversity, we plot validation LPIPS against mean pairwise LPIPS across training checkpoints. Each point corresponds to one checkpoint and is placed in the two-dimensional plane according to:

- **LPIPS to the reference HR image** (fidelity),
- **mean pairwise LPIPS between samples** (diversity).

Points are connected in chronological order, so that the resulting curve represents the trajectory of the model through the fidelity–diversity plane during training.

This visualization is particularly useful for stochastic SR, because it shows not only the final performance but also how the model moves between low-diversity and high-diversity regimes. In this representation, lower LPIPS corresponds to better perceptual fidelity, while higher pairwise LPIPS corresponds to stronger diversity. Hence, the desirable region is the upper-left part of the plane.

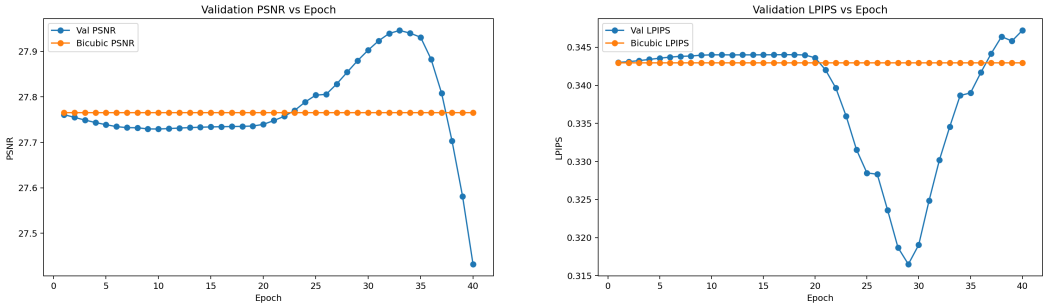
In our experiments, the noise-conditioned baseline tends to remain in a relatively low-diversity region, which indicates that noise injection alone is insufficient to induce a strong one-to-many effect. The proposed drifting model, on the contrary, moves toward substantially higher pairwise LPIPS values while maintaining competitive LPIPS to the reference HR image. This behavior is consistent with the intended role of conditional drifting: to increase diversity within the admissible solution set rather than to produce uncontrolled deviations from the measurement.

4.5 Learning-curve analysis

In addition to the fidelity–diversity trajectory, we monitor the evolution of the main validation metrics over training epochs. In particular, we record validation PSNR, validation LPIPS, and pairwise diversity metrics for the noise-conditioned baseline and the proposed drifting model.

This learning-curve analysis serves as the main error-analysis component of the computational experiment. It allows us to examine how the optimization process is reflected in the external quality criteria and whether the best perceptual checkpoint coincides with the last training epoch. In our experiments, these points may differ substantially, which justifies validation-based checkpoint selection.

Figure 2: Trajectory in the fidelity–diversity plane during training. Each point corresponds to one checkpoint and is plotted by validation LPIPS and mean pairwise LPIPS. Points are connected in training order. The noise-conditioned baseline and the proposed drifting model occupy different regions of this plane.



(a) PSNR

(b) LPIPS

Figure 3: Learning curves of the proposed drifting model evaluated on the first 20 images of the validation set.

4.6 Ablation on the residual feature encoder

The proposed method computes drifting in a residual feature space, so the choice of encoder used for the drifting loss is an important design variable. Table 3 compares several encoder choices or feature-family variants.

This experiment addresses whether the observed improvement is tied to one particular feature representation or whether the drifting mechanism is robust across reasonable encoder choices. Since the original Drifting framework also emphasizes the importance of feature-space drifting and multi-feature constructions, this comparison is particularly relevant for our conditional SR adaptation.

4.7 Ablation on the number of drifting feature levels

The proposed method supports drifting losses computed across several feature levels. To evaluate the effect of multi-scale residual drifting, we compare one-level, two-level, and three-level variants in Table 4.

This ablation tests whether richer multi-scale residual geometry improves the fidelity–diversity trade-off. A single-level drifting loss constrains the model only at one resolution scale, whereas multi-level drifting can simultaneously regularize coarse structures and fine details.

4.8 Sensitivity to the repulsion strength

Although the theoretically clean variant of the drifting field uses a symmetric attraction–repulsion form, in practice it is useful to analyze the sensitivity of the method to the strength of the same-condition repulsion term. To this end, we introduce a scalar coefficient λ_{neg} that scales the repulsive component while keeping all other parts of the training protocol fixed.

Concretely, in this experiment we use the conditional drifting field in the form

$$V^{(k)} = V^{\text{pos},(k)} - \lambda_{\text{neg}} V^{\text{same},(k)}.$$

We compare several values of λ_{neg} in order to study the trade-off between fidelity and diversity. Intuitively, smaller values of λ_{neg} should yield weaker separation between samples generated from the same LR input, whereas larger values should encourage stronger diversity but may also harm reconstruction fidelity if the repulsion becomes too strong.

In our setting, increasing λ_{neg} generally increases inter-sample diversity, as reflected by pairwise LPIPS, but excessively strong repulsion may reduce reconstruction fidelity. This confirms that the

Table 3: Comparison of feature encoders or feature-family variants used for the drifting loss.

Encoder / feature family	PSNR ↑	LPIPS ↓	Pair LPIPS ↑
VGG, 1 level	XX.XX	XX.XXXX	XX.XXXX
VGG, 2 levels	XX.XX	XX.XXXX	XX.XXXX
VGG, 3 levels	XX.XX	XX.XXXX	XX.XXXX
Encoder B	XX.XX	XX.XXXX	XX.XXXX
Encoder C	XX.XX	XX.XXXX	XX.XXXX

Table 4: Ablation on the number of drifting feature levels.

Feature levels	PSNR ↑	LPIPS ↓	Pair LPIPS ↑
1 level	XX.XX	XX.XXXX	XX.XXXX
2 levels	XX.XX	XX.XXXX	XX.XXXX
3 levels	XX.XX	XX.XXXX	XX.XXXX

repulsion coefficient controls a meaningful fidelity–diversity trade-off and should be chosen jointly with the overall drifting strength.

4.9 Ablation on the admissible positive set

To evaluate the effect of the admissible positive set, we compare a variant that uses only the ground-truth HR target ($M = 1$) with a variant that uses multiple admissible positives ($M = 4$), including mild sharpened or high-pass-enhanced HR variants that remain consistent with the same LR observation. Results are shown in Table 6.

This experiment directly tests one of the core modeling assumptions of the paper. If multiple admissible positives improve diversity without severely degrading fidelity, this supports the claim that one-to-many SR benefits from target sets that better reflect conditional ambiguity than a single HR target alone.

4.10 Ablation on generator architecture

To ensure that the observed gains are not tied to one particular generator design, we additionally compare several generator configurations in Table 7. In all cases, we keep the training objective fixed and vary only the structure of the SR backbone.

This experiment addresses whether the benefits of conditional drifting persist across different model capacities. If the same trend is observed for several architectures, this would indicate that the proposed drifting objective is not specific to one backbone.

4.11 Ablation on noise injection

Since the proposed model is stochastic, the way noise is injected into the generator is itself a meaningful design choice. Table 8 compares several noise-injection strategies, such as spatial noise only, global noise only, and the joint use of both.

This experiment helps determine whether diversity is primarily controlled by global latent variation, by spatial perturbation, or by their combination. It also clarifies whether the drifting objective can effectively exploit different types of stochastic conditioning.

4.12 Qualitative comparison

In addition to quantitative metrics, we compare visual samples produced by the noise-conditioned baseline and the proposed drifting model for the same LR input under different noise realizations. Figure 4 illustrates two aspects of interest: (i) fidelity to the global LR structure, and (ii) variability of local high-frequency details across samples.

Table 5: Sensitivity analysis with respect to the same-condition repulsion strength λ_{neg}.

λ_{neg}	PSNR ↑	LPIPS ↓	Pair LPIPS ↑
0.00	XX.XX	XX.XXXX	XX.XXXX
0.25	XX.XX	XX.XXXX	XX.XXXX
0.50	XX.XX	XX.XXXX	XX.XXXX
0.75	XX.XX	XX.XXXX	XX.XXXX
1.00	XX.XX	XX.XXXX	XX.XXXX

Table 6: Ablation on the size of the admissible positive set.

Number of positives M	PSNR ↑	LPIPS ↓	Pair LPIPS ↑
1	XX.XX	XX.XXXX	XX.XXXX
4	XX.XX	XX.XXXX	XX.XXXX

Visually, the noise-conditioned baseline often produces only weak variation between samples, even when noise is present. By contrast, the proposed drifting model tends to generate a broader family of plausible local textures while preserving the same large-scale image structure. This behavior is consistent with the goal of conditional one-to-many super-resolution.

4.13 Discussion

The experiments suggest that stochastic conditioning alone is not sufficient to produce a strong one-to-many SR effect. Although a noise-conditioned baseline can improve reconstruction quality, the resulting predictions remain relatively close to each other. The proposed drifting mechanism changes this behavior by explicitly encouraging motion toward admissible HR residual configurations while repelling competing solutions generated for the same LR condition.

At the same time, the results indicate that model selection is important: the best perceptual checkpoint and the last checkpoint may differ substantially. This is expected in a stochastic perceptual setting, where the trade-off between fidelity and diversity evolves during training. Consequently, validation-based checkpointing is essential for a fair comparison between stochastic SR models.

5 Conclusion

We presented a conditional adaptation of Generative Drifting to one-step one-to-many super-resolution. The proposed method treats SR as an observation-conditioned generative problem and performs drifting in a residual feature space relative to the bicubic baseline. In this formulation, generated samples are attracted toward admissible HR residual configurations associated with the same LR input while being discouraged from collapsing to identical reconstructions.

The resulting framework preserves one-step inference and explicitly addresses the ambiguity of super-resolution at training time rather than suppressing it through deterministic regression. Our experiments indicate that conditional drifting can substantially increase inter-sample diversity while maintaining competitive reconstruction fidelity. This suggests that drifting provides a promising mechanism for controlled diversity in super-resolution.

Several directions remain open for future work. First, richer feature encoders and stronger feature families may further improve the fidelity–diversity trade-off. Second, the design of admissible positive sets can be extended beyond mild HR transformations. Third, the interaction between generator architecture, noise injection, and drifting dynamics deserves a more systematic study. More broadly, the present work suggests that training-time distribution evolution may be a useful principle not only for unconditional generation but also for ill-posed conditional restoration problems.

Table 7: Ablation on generator architecture.

Generator variant	PSNR ↑	LPIPS ↓	Pair LPIPS ↑
Generator A	XX.XX	XX.XXXX	XX.XXXX
Generator B	XX.XX	XX.XXXX	XX.XXXX
Generator C	XX.XX	XX.XXXX	XX.XXXX

Table 8: Ablation on noise injection strategy.

Noise injection	PSNR ↑	LPIPS ↓	Pair LPIPS ↑
Global noise only	XX.XX	XX.XXXX	XX.XXXX
Spatial noise only	XX.XX	XX.XXXX	XX.XXXX
Global + spatial noise	XX.XX	XX.XXXX	XX.XXXX

A Additional implementation details

A.1 Feature family details

The feature family $\Phi = \{\phi_j\}_{j=1}^{J}$ is constructed from intermediate feature maps of a fixed encoder. Let

$$F_1(x), F_2(x), \ldots, F_S(x), \qquad F_s(x) \in \mathbb{R}^{C_s \times H_s \times W_s},$$

denote the feature maps extracted at different encoder stages.

For each stage s, we include local descriptors at spatial locations (a, b):

$$\phi_{s,a,b}(x) = F_s(x)[:, a, b] \in \mathbb{R}^{C_s}, \qquad 1 \le a \le H_s,\ 1 \le b \le W_s.$$

To incorporate slightly larger spatial support, we may also include pooled regional descriptors. For a pooling factor ρ, let

$$F_s^{(\rho)}(x) = \mathrm{Pool}_\rho(F_s(x)),$$

where Pool_ρ denotes average pooling over $\rho \times \rho$ neighborhoods, and define

$$\phi_{s,\alpha,\beta}^{(\rho)}(x) = F_s^{(\rho)}(x)[:, \alpha, \beta] \in \mathbb{R}^{C_s}.$$

The complete feature family is obtained by collecting all selected descriptors across stages, spatial locations, and pooling levels:

$$\Phi = \{\phi_j\}_{j=1}^{J}.$$

For each generated sample $\hat{x}_i^{(k)}$ and admissible positive target $x_{i,m}^{+}$, residual feature vectors are computed relative to the bicubic baseline $x_i^{UP} = u(x_i^{LR})$ as

$$g_{i,j}^{(k)} = \phi_j(\hat{x}_i^{(k)}) - \phi_j(x_i^{UP}), \qquad p_{i,j}^{(m)} = \phi_j(x_{i,m}^{+}) - \phi_j(x_i^{UP}).$$

In practice, this construction allows the drifting loss to compare generated and admissible target residuals across multiple spatial scales while keeping the low-frequency bicubic structure factored out.

A.2 Auxiliary negatives

In addition to same-condition generated negatives, the repulsion term may optionally include auxiliary negatives. In our setting, we consider two types of such negatives: cross-condition negatives collected from other LR conditions, and hard negatives constructed from mildly blurred variants of the paired HR image.

Let

$$\mathcal{Q}_j = \{q_{j,\ell}\}_{\ell=1}^{L_Q}, \qquad q_{j,\ell} \in \mathbb{R}^{C_j},$$

Figure 4: Qualitative comparison of multiple samples generated for the same LR input. Each row corresponds to one LR image and shows bicubic upsampling, the noise-conditioned baseline samples, and the proposed drifting samples under different noise seeds.

denote a set of cross-condition residual feature vectors, obtained either from other samples in the current mini-batch or from a memory queue.

We also define hard negatives from blurred HR variants:

$$x_{i,h}^{-} = \mathcal{B}_{\rho_h}(x_i^{HR}), \qquad h = 1, \ldots, H,$$

where $\mathcal{B}_{\rho_h}$ denotes a mild blur transformation with parameter ρ_h. Their residual feature vectors are

$$h_{i,j}^{(h)} = \phi_j(x_{i,h}^{-}) - \phi_j(x_i^{UP}).$$

When auxiliary negatives are used, the full repulsion term becomes

$$V_{i,j}^{\mathrm{neg},(k)} = V_{i,j}^{\mathrm{same},(k)} + V_{i,j}^{\mathrm{queue},(k)} + V_{i,j}^{\mathrm{hard},(k)},$$

where

$$V_{i,j}^{\mathrm{queue},(k)} = \lambda_{\mathrm{queue}} \sum_{\ell=1}^{L_Q} c_{i,j,\ell}^{(k)} \big(q_{j,\ell} - g_{i,j}^{(k)}\big),$$

and

$$V_{i,j}^{\mathrm{hard},(k)} = \lambda_{\mathrm{hard}} \sum_{h=1}^{H} d_{i,j,h}^{(k)} \big(h_{i,j}^{(h)} - g_{i,j}^{(k)}\big).$$

The corresponding weights are distance-based and normalized within each auxiliary set:

$$c_{i,j,\ell}^{(k)} \propto \exp\left(-\frac{\|g_{i,j}^{(k)} - q_{j,\ell}\|_2}{\tau_{\mathrm{queue}}}\right), \qquad \sum_{\ell=1}^{L_Q} c_{i,j,\ell}^{(k)} = 1,$$

and

$$d_{i,j,h}^{(k)} \propto \exp\left(-\frac{\|g_{i,j}^{(k)} - h_{i,j}^{(h)}\|_2}{\tau_{\mathrm{hard}}}\right), \qquad \sum_{h=1}^{H} d_{i,j,h}^{(k)} = 1.$$

These auxiliary negatives are not essential to the basic conditional drifting mechanism, but they can further discourage oversmoothed solutions and improve separation from feature configurations that are either cross-condition inconsistent or overly close to blurred HR residuals.

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